Forecasting Methods at a Glance

SES: Simple Exponential Smoothing

$$\hat{y}_t = \alpha y_{t-1} + (1-\alpha)\hat{y}_{t-1}$$

Prediction for period t | Observed value of previous period | Prediction for previous period

Already considers all past observations, with **exponentially** decreasing weights:

$$\hat{y}_t = \alpha y_{t-1} + (1-\alpha)[\alpha y_{t-2} + (1-\alpha)\hat{y}_{t-2}]$$
$$= \alpha y_{t-1} + \alpha(1-\alpha) y_{t-2} + (1-\alpha)^2 \hat{y}_{t-2}$$
$$\hat{y}_t = \alpha y_{t-1} + \alpha(1-\alpha) y_{t-2} + (1-\alpha)^2 [\alpha y_{t-3} + (1-\alpha)\hat{y}_{t-3}]$$
$$= \alpha y_{t-1} + \alpha(1-\alpha) y_{t-2} + \alpha(1-\alpha)^2 y_{t-3} + (1-\alpha)^3 \hat{y}_{t-3}$$

Croston: Intermittent Demand Forecasting

Designed for **intermittent** demand: many periods have zero demand, occasional periods have positive demand.

e.g.

Week	1	2	3	4	5	6	7	8
Sales	0	5	0	0	3	0	0	4

Prediction for period t $\hat{y}_t = \frac{\hat{q}_t}{\hat{p}_t}$ — $\hat{q}_t$: Estimated demand size when demand occurs; $\hat{p}_t$: Estimated number of periods between nonzero demands

At week 2 (initialization): $\hat{y}_2 = \frac{5}{2} = 2.5\ sales/week$

At week 3 and 4 (no demand): $\hat{y}_3 = \hat{y}_4 = 2.5\ sales/week$

Only updates at nonzero values

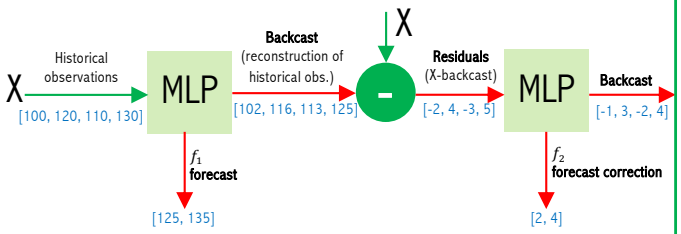
At week 5 (new observed demand): $say\ \alpha = 0.3$

$$\hat{y}_5 = \frac{\hat{q}_5 = \alpha y_5 + (1-\alpha)\hat{q}_4}{\hat{p}_t = \alpha r_t + (1-\alpha)\hat{p}_4}$$
$$= \frac{\hat{q}_5 = 0.3(3) + (1-0.3)5}{\hat{p}_t = 0.3(3) + (1-0.3)\,2}$$
$$= \frac{4.4}{2.3} = 1.91\ sales/week$$

Forecast are updated **after** observing demand at time t.

N-BEATS

Can predict multiple future time steps simultaneously



Final forecast $\hat{y} = f_1 + f_2$

=[125, 135]+[2, 4]

=[127, 139]

Second block can contribute corrections based on the remaining unexplained information: uses the residual to improve the final forecast

ETS: Error, Trend, Seasonality

h : # periods into the future

$$\hat{y}_{t+h|t} = l_t + hb_t + s_{t+h-m(k+1)}$$

Prediction for period (t+h) | Level at time t | Trend at time t | Seasonal adjustment for the corresponding period

Level l_t : Smoothed estimate

$$l_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1})$$

Observed value at time t minus previous seasonal adjustment | Previous level | Previous trend

e.g. $l_t = 0.5(150 - 30) + (1-0.5)(100 + 10) = 115$

Trend b_t : How quickly values are increasing or decreasing

$$b_t = \beta(l_t - l_{t-1}) + (1-\beta)(b_{t-1})$$

Change between current and prev. levels | Previous trend

e.g. $b_t = 0.3(115 - 100) + (1-0.3)(10) = 11.5$

Seasonality s : Seasonal adjustment

m: length of seasonal cycle

k: $\lfloor (h-1)/m \rfloor$: determines which stored seasonal index to reuse

ARIMA: AutoRegressive Integrated Moving Average

Prediction for period t

$$\hat{y}_t = y_{t-1} + \hat{z}_t$$

Observed value of previous period | Predicted change from t-1 to t

For ARIMA(p=2, d=1, q=1):

$$\hat{z}_t = c + \emptyset_1 z_{t-1} + \emptyset_2 z_{t-2} + \theta_1 \varepsilon_{t-1}$$

Intercept | AR: Previous changes | MA: Previous prediction error

I - Integrated (differencing) : Because d=1, calculate changes between *consecutive* observations: $z_t = y_t - y_{t-1}$

AR - Autoregressive (AR): Use previous changes to predict the next change

$$AR_t = \emptyset_1 z_{t-1} + \emptyset_2 z_{t-2}$$

e.g. $AR_t = 0.7(20) + 0.2(10) = 16$

MA - Moving Average: Use previous prediction errors to adjust the prediction

$$MA_t = \theta_1 \varepsilon_{t-1}$$

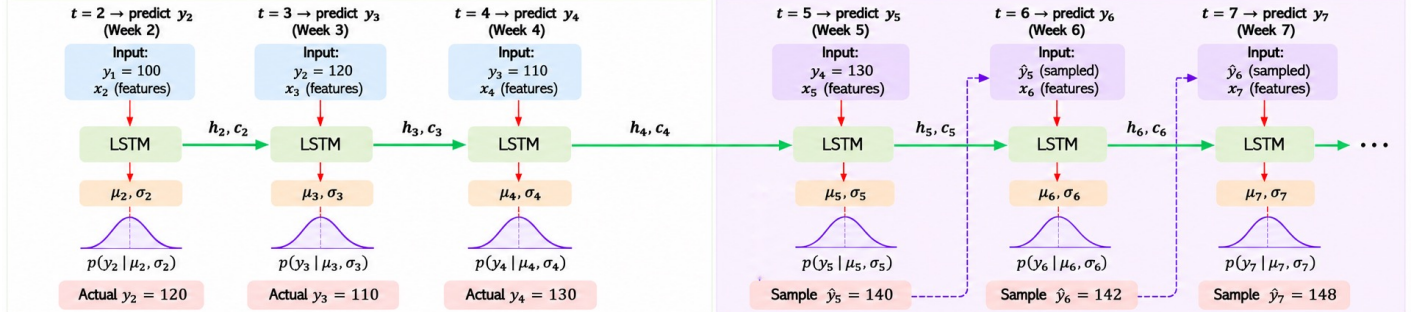
$$\varepsilon_{t-1} = z_{t-1} - \hat{z}_{t-1}$$

Actual change observed at time t-1 | Predicted change observed at time t-1

$c, \emptyset_1, \emptyset_2, \theta_1$ are learned

DeepAR

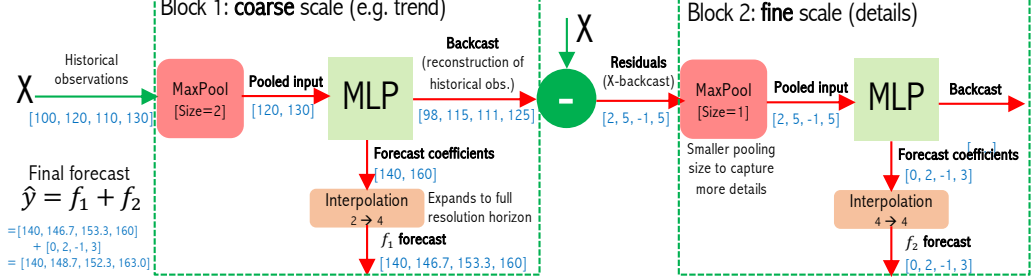
At each time step, it outputs the parameters of a probability distribution for the next value. Input features are known: time features, promotions, category, etc.



N-HITS

Designed for forecasting long horizons.

Can predict multiple future time steps simultaneously



Final forecast $\hat{y} = f_1 + f_2$

=[140, 146.7, 153.3, 160]

+ [0, 2, -1, 3]

= [140, 148.7, 152.3, 163.0]

Statistical methods

Deep learning - Point forecasting

Deep learning - Probabilistic forecasting

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v.1.0